NOTING regulation 13 of MARPOL Annex VI, which makes the NO x Technical Code 2008 mandatory under that Annex,
NOTING ALSO that the use of NO x-reducing devices is envisaged in the NO X Technical

---CHUNK---

SOURCE: MEPC.399(83).pdf
NOTING regulation 13 of MARPOL Annex VI, which makes the NO x Technical Code 2008 mandatory under that Annex,
NOTING ALSO that the use of NO x-reducing devices is envisaged in the NO X Technical Code 2008 and that selective catalytic reduction (SCR) systems are such NO x-reducing devices for compliance with the Tier III NOX limit,
NOTING FURTHER that, at its seventy-first session, it adopted, by resolution MEPC.291(71), the 2017 Guidelines addressing additional aspects to the NOX Technical Code 2008 with regard to particular requirements related to marine diesel engines fitted with Selective Catalytic Reduction (SCR) Systems (2017 SCR Guidelines), and, at its seventy -fourth session, by resolution MEPC.313(74), amendments thereto,
RECOGNIZING the need to update the 2017 SCR Guidelines in line with the latest developments,
HAVING CONSIDERED, at its eighty -third session, a draft revision of the 2017 SCR Guidelines, prepared by the Sub-Committee on Pollution Prevention and Response,
1 ADOPTS the 2025 Guidelines on selective catalytic reduction (SCR) systems, as set out in the annex to the present resolution;

---CHUNK---

SOURCE: MEPC.399(83).pdf
Guidelines, prepared by the Sub-Committee on Pollution Prevention and Response,
1 ADOPTS the 2025 Guidelines on selective catalytic reduction (SCR) systems, as set out in the annex to the present resolution;
2 INVITES Administrations to implement the 2025 SCR Guidelines and apply them to SCR systems installed on ships the keels of which are laid or which are at a similar stage of construction on or after 1 November 2025; or SCR systems installed on ships the kee ls of which are laid or which are at a similar stage of construction before 1 November 2025 which have a contractual delivery date of SCR systems to the ship on or after 1 May 2026 or, in the absence of a contractual delivery date, the actual delivery of the SCR system to the ship on or after 1 May 2026;
3 REQUESTS Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of shipowners, ship operators, shipbuilders, marine diesel engine manufacturers and any other interested parties;
4 AGREES to keep these Guidelines under review in light of experience gained with their application, with a view to incorporating them into the NOx Technical Code 2008;

---CHUNK---

SOURCE: MEPC.399(83).pdf
diesel engine manufacturers and any other interested parties;
4 AGREES to keep these Guidelines under review in light of experience gained with their application, with a view to incorporating them into the NOx Technical Code 2008;
5 ALSO AGREES that these Guidelines supersede the 2017 SCR Guidelines, adopted by resolution MEPC.291(71) and amended by resolution MEPC.313(74).
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 2

I:\MEPC\MEPC 83-17-Add.1.docx
ANNEX
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMS
TABLE OF CONTENTS
1 INTRODUCTION
2 GENERAL
2.1 Purpose
2.2 Application
2.3 Definitions
3 PRE-CERTIFICATION PROCEDURE
3.1 General
3.2 Technical File and onboard NOx verification procedures
3.3 Measures to minimize reductant slip
3.4 Pre-certification procedure
3.5 EIAPP certificate
4 FAMILY AND GROUP CONCEPTS FOR ENGINE SYSTEMS FITTED WITH SCR
5 TEST PROCEDURES FOR SCHEME A
5.1 General
5.2 Calculation of gaseous emissions
6 TEST PROCEDURES FOR SCHEME B
6.1 General
6.2 Verification test procedures for an engine

---CHUNK---

SOURCE: MEPC.399(83).pdf
4 FAMILY AND GROUP CONCEPTS FOR ENGINE SYSTEMS FITTED WITH SCR
5 TEST PROCEDURES FOR SCHEME A
5.1 General
5.2 Calculation of gaseous emissions
6 TEST PROCEDURES FOR SCHEME B
6.1 General
6.2 Verification test procedures for an engine
6.3 Test procedures for SCR chambers
6.4 Calculation of the specific emission
6.5 Test report to be submitted to the Administration
7 ONBOARD CONFIRMATION TEST FOR SCHEME B
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 3
I:\MEPC\MEPC 83-17-Add.1.docx
1 INTRODUCTION
1.1 The use of NOx-reducing devices is envisaged in section 2.2.5 of the NO x Technical Code 2008 (NTC 2008) and a Selective Catalytic Reduction (SCR) system is one of such devices.
1.2 The NTC 2008 contains two ways for pre -certification of engine systems fitted with NOx-reducing devices:
.1 engine fitted with SCR: approval in accordance with paragraph 2.2.5.1 and test in accordance with chapter 5 of the NTC 2008; and
.2 a simplified measurement method in accordance with section 6.3 of the NTC 2008 as regulated in paragraph 2.2.5.2 (Primary failure case) of the Code.

---CHUNK---

SOURCE: MEPC.399(83).pdf
test in accordance with chapter 5 of the NTC 2008; and
.2 a simplified measurement method in accordance with section 6.3 of the NTC 2008 as regulated in paragraph 2.2.5.2 (Primary failure case) of the Code.
1.3 According to paragraph 2.2.5.1 of the NTC 2008, where a NOx-reducing device is to be included within the EIAPP certification, it must be recognized as a component of the engine, and its presence shall be recorded in the engine's Technical File.
1.4 Administrations are invited to take these Guidelines into account when certifying engines fitted with SCR.
2 GENERAL
2.1 Purpose
The purpose of these Guidelines is to provide guidance in addition to the requirements of the NTC 2008 for design, testing, surveys and certification of marine diesel engines fitted with an SCR system to ensure its compliance with the requirements of regulation 13 of MARPOL Annex VI.
2.2 Application
These Guidelines apply to marine diesel engines fitted with SCR for compliance with regulation 13 of MARPOL Annex VI.
2.3 Definitions
Unless provided otherwise, the terms in these Guidelines have the same meaning as the terms

---CHUNK---

SOURCE: MEPC.399(83).pdf
2.2 Application
These Guidelines apply to marine diesel engines fitted with SCR for compliance with regulation 13 of MARPOL Annex VI.
2.3 Definitions
Unless provided otherwise, the terms in these Guidelines have the same meaning as the terms defined in regulation 2 of MARPOL Annex VI and in section 1.3 of the NTC 2008.
2.3.1 "Engine system fitted with SCR" means a system consisting of a marine diesel engine, an SCR chamber and a reductant injection system. When a control device on NO x-reducing performance is provided, it is also regarded as a part of the system.
2.3.2 "Catalyst block" means a block of certain dimension through which exhaust gas passes and which contains catalyst composition on its inside surface to reduce NO x from the exhaust gas.
2.3.3 "SCR chamber" means an integrated unit which contains the catalyst block(s) and into which flow exhaust gas and reductant.

2.3.4 "Reductant injection system" means a system which consists of the pump(s) to supply

---CHUNK---

SOURCE: MEPC.399(83).pdf
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 4
I:\MEPC\MEPC 83-17-Add.1.docx
2.3.4 "Reductant injection system" means a system which consists of the pump(s) to supply reductant to the nozzle(s), the nozzle(s) spraying reductant into the exhaust gas stream and control device(s) of the spray.
2.3.5 "AV (area velocity) value" means a value of the exhaust gas flow rate passing through

the catalyst blocks (m 3/h) per total active surface area of the catalyst blocks in the SCR chamber (m2). Therefore, the unit of AV value is (m/h). The exhaust gas flow volume is the volume defined at 0°C and 101.3 kPa.
2.3.6 "SV (space velocity) value" means a value of the exhaust gas flow rate passing through the catalyst block(s) (m 3/h) per total volume of the catalyst block(s) in the SCR chamber (m 3). Therefore, the unit of SV value is (1/h). The exhaust gas flow volume is the volume defined at 0°C and 101.3 kPa.
2.3.7 "Total volume of the catalyst block" means the volume (m 3) based on the outer dimensions of the catalyst block.
2.3.8 "LV (linear velocity) value" means a value of the exhaust gas flow rate passing through

---CHUNK---

SOURCE: MEPC.399(83).pdf
2.3.7 "Total volume of the catalyst block" means the volume (m 3) based on the outer dimensions of the catalyst block.
2.3.8 "LV (linear velocity) value" means a value of the exhaust gas flow rate passing through the catalyst blocks (m3/h) per catalyst block's section (m2) in a normal direction of exhaust gas flow. Therefore, the unit of LV value is (m/h). The exhaust gas flow volume is the volume defined at 0°C and 101.3 kPa.
2.3.9 "Block section" means the cross -sectional area (m 2) of the catalyst block based on the outer dimensions.
2.3.10 "NOx reduction rate " means a value deriving from the following formula. Unit of